Theory & Manual

for

AlkCalc

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I. Introduction

Bound single-electron eigenfunctions of alkali-metal atoms and alkaline-earth metal ions are of significant relevance in quantum computing, quantum information processing, quantum simulation, Rydberg physics, quantum optics, ... However, expect for the special case of the Hydrogen atom, it is not possible to solve the time-independent Schrödinger equation analytically. Because of this, there is the need to obtain eigenenergies and eigenstates numerically. The library `AlkCalc` is designed for exactly this purpose: it allows the user to compute the relevant part of the bound-state energy spectrum with the associated eigenstates.

The software `AlkCalc` is split into two parts: a first part used to compute the eigenenergies and eigenstates *once* and store the resulting data. And a second part, referred to as the *library functions*, which implement useful functionality for reading and processing the precomputed atomic data. For instance, there is as function to compute radial transition matrix elements.

In the various README files included in the file tree of `AlkCalc` , technical details for computing the eigenenergies and eigenstates are provided. However, these files do not include a discussion on the theory and physics behind the code. Further, there is no reference manual for the library functions.

This documents purpose is to add the missing pieces: in Sec. II the theory and the physics associated with the code are discussed. Section III is a comprehensive reference manual for the library functions of `AlkCalc` .

II. Theory

This section is organised as follows: in Subsec. i the theory and the physics of single-atom/ion systems are discussed. In Subsec. ii the numerical method based on B-splines is presented. In Subsec. iii it is shown how radial transitions matrix elements are computed by `AlkCalc` . In Subsec. iv it is shown how oscillator strengths for electronic transitions are computed. Finally, in Subsec. iv it is shown how lifetimes of electronic states are computed by `AlkCalc` .

i. Single-atom/ion theory

The starting point is an alkali-metal atom or alkaline-earth-metal ion. We are interested in systems where all electrons are in closed shells except for one valence electron. Examples of such systems are ^{1}H, ^{38}Rb, ^{88}Sr^{+}. This property allows a Kepler-problem description for the 'effective nucleus' (the nucleus 'screened' by the closed-shell electrons) and the valence electron. The dynamics of the valence electron and the effective nucleus can be described with the following Hamiltonian:

$$H = \frac{\boldsymbol{p}^2}{2m} + V(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2)\,. \tag{II.1}$$

Here, $m = M m_{\mathrm{e}}/(M + m_{\mathrm{e}})$ is the reduced mass of the system, composed of the electron's mass m_{e} and the atom's or ion's mass M. The centre-of-mass motion is decoupled and $\boldsymbol{r}$, $\boldsymbol{p}$ are the relative canonical co-ordinates, i.e., $\boldsymbol{r}$ is the distance vector between the centre-of-mass position of the effective nucleus and the position of the valence electron. Further, $\boldsymbol{L}$, $\boldsymbol{S}$, and $\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$ are the orbital angular momentum operator, the electron's spin operator, and the total angular momentum operator, respectively. Finally, the radial potential V, depending on the distance $r = \|\boldsymbol{r}\|$, describes the interaction between the valence electron and the effective nucleus. It accounts for several effects: far from the nucleus the valence electron experiences a Coulomb potential with effective charge $Z_{\mathrm{eff}} = 1$ for alkali-metal atom and $Z_{\mathrm{eff}} = 2$ for singly ionised alkaline-earth metal atoms. This is because the nucleus of charge Z is 'screened' by the $Z-1$ (or $Z-2$, for alkaline-earth metal ions) ground-state electrons in closed-shells. However, valence electrons with orbits sufficiently close to the nucleus experience the non-trivial charge distribution posed by the nucleus in conjugation with the tightly bound electron cloud. This leads to short-ranged corrections to the effective Coulomb potential with charge Z_{eff} at long distances. Following Refs. [1–4] these corrections are described by the following potential:

$$V(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2) = V_{\mathrm{C}}(r) + V_{\mathrm{P}}(r) + V_{\mathrm{R}}(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2)\,. \tag{II.2}$$

The first contribution is

$$V_{\mathrm{C}}(r) = -\frac{e^2}{4\pi\varepsilon_0}\frac{Z_{\mathrm{n}}(r)}{r}\,,\ Z_{\mathrm{n}}(r) = Z_{\mathrm{c}} + (Z - Z_{\mathrm{c}})\mathrm{e}^{-k_1 r} + k_2 r \mathrm{e}^{-k_3 r} - k_4 r^2 \mathrm{e}^{-k_3 r}\,. \tag{II.3}$$

This is a modified Coulomb potential which has the tail of a Coulomb potential with charge Z_{eff} and short-ranged corrections for small radii accounting for the charge-distribution of the effective nucleus. Here, the parameters k_j ($j = 1, 2, 3, 4$) are empirically determined from spectroscopic data [1, 2]. The second contribution to the potential (II.2) is

$$V_{\mathrm{P}}(r) = -\frac{e^2}{(4\pi\varepsilon_0)^2}\frac{\alpha_{\mathrm{d}}}{2r^4}\left[1 - \exp\left(-\frac{r^6}{r_{\mathrm{c}}^6}\right)\right]. \tag{II.4}$$

It accounts for the polarisability of the screened nucleus, which gets polarised by the presence of the valence electron. Here, r_{c} is the effective nucleus' size and α_{d} its static electric polarisability. Finally,

$$V_{\mathrm{R}}(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2) = \frac{1}{2m^2c^2}\frac{V'_{\mathrm{NR}}(r)}{rN(r)}\boldsymbol{L}\cdot\boldsymbol{S}\,,\ N(r) = \left[1 - \frac{V_{\mathrm{NR}}(r)}{2mc^2}\right]^2\,,\ V_{\mathrm{NR}}(r) = V_{\mathrm{C}}(r) + V_{\mathrm{P}}(r)\,. \tag{II.5}$$

It describes (relativistic) coupling between the orbital angular momentum and the valence electrons' spin. This expression results from relativistic perturbation theory [3]. Here, V_{NR} is the non-relativistic model potential and c the speed of light. In Ref. [4] the potential (II.5) is employed for the treatment of alkaline-earth metal ions. In Ref. [5] the simplified version

$$V_{\mathrm{R}}(r) = \frac{\alpha}{2r^3}\boldsymbol{L}\cdot\boldsymbol{S} \tag{II.6}$$

is employed for alkali-metal atoms. Here, α is the fine-structure constant. Throughout `AlkCalc` relativistically correct version in Eq. (II.5), not the simplified version in Eq. II.6, is employed. Finally, for expressing the spin-operator $\boldsymbol{S}$ in Eq. (II.5) and Eq. (II.6) the representation in Pauli's (spin) matrices is chosen:

$$\boldsymbol{S} = \frac{\hbar}{2}\begin{bmatrix}\sigma_x\\ \sigma_y\\ \sigma_z\end{bmatrix}. \tag{II.7}$$

The Pauli spin matrices are

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \ \sigma_y = \begin{bmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{bmatrix}, \ \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \tag{II.8}$$

This simple choice for representing the spin degreed of freedom is possible because throughout this work the interest is in the case where the total spin is $s = 1/2$.

At this point the Hamiltonian which determines the eigenenergies and the eigenstates is defined. In the following, the goal is to cast eigenvalue problem posed by Hamiltonian (II.1) into a more practical form for the numerical procedure. The first step is to rewrite the product $\boldsymbol{L} \cdot \boldsymbol{S}$ in terms of the total angular momentum operator $\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$:

$$\boldsymbol{L} \cdot \boldsymbol{S} = \frac{1}{2}\left[\boldsymbol{J}^2 - \boldsymbol{L}^2 - \boldsymbol{S}^2\right]. \tag{II.9}$$

Next, the square of the momentum operator in spherical co-ordinates is rewritten:

$$\boldsymbol{p}^2 = (-\mathrm{i}\hbar\nabla)^2 = -\hbar^2\nabla^2 = -\frac{\hbar^2}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\boldsymbol{L}^2}{r^2}. \tag{II.10}$$

The second contribution is the centrifugal barrier. In a classical Kepler problem this is what prevents the electron crashing into the nucleus. Expressed in the spherical coordinates, Hamiltonian (II.1) takes the form

$$H = -\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + V_{\text{eff}}(r;\boldsymbol{L}^2,\boldsymbol{S}^2,\boldsymbol{J}^2), \ V_{\text{eff}}(r;\boldsymbol{L}^2,\boldsymbol{S}^2,\boldsymbol{J}^2) = V(r;\boldsymbol{L}^2,\boldsymbol{S}^2,\boldsymbol{J}^2) + \frac{\boldsymbol{L}^2}{2mr^2}. \tag{II.11}$$

The full Hilbert space this Hamiltonian acts on can be split into an angular momentum factor and a radial one. A basis of the angular momentum Hilbert space is given by the simultaneous eigenstates Φ_{l,s,j,m_j} of the operators (Cartan subalgebra) $\boldsymbol{L}^2$, $\boldsymbol{S}^2$, $\boldsymbol{J}^2$, and J_z. Note that a simultaneous eigenbasis of $\boldsymbol{L}^2$, $\boldsymbol{S}^2$, $\boldsymbol{J}^2$, and J_z exists, because these operators commute pairwise. The so-called *spinors* $\boldsymbol{\Phi}_{l,s,j,m_j}$ constitute the so-called *coupled basis*, also known as the *fine-structure* basis in the literature. These spinors are given by linear combinations of the uncoupled basis states

$$Y_{l,m_l} \otimes \boldsymbol{\chi}_{s,m_s} \tag{II.12}$$

via SU(2) Clebsch-Gordan coefficients [6]. Here, $Y_{l,m_l} : [0,\pi]\times[0,2\pi] \to \mathbb{C}$ are spherical harmonics defined as

$$Y_{l,m_l}(\vartheta,\varphi) = \sqrt{\frac{2l+1}{4\pi}\frac{(l-m_l)!}{(l+m_l)!}}P_l^{m_l}(\cos(\vartheta))\,\mathrm{e}^{\mathrm{i}m_l\varphi}. \tag{II.13}$$

In this expression $P_l^{m_l} : [-1,1] \to \mathbb{R}$ are the associated Legendre polynomials defined for $l = 0,1,2,\ldots$ and $m_l = 0,1,\ldots,l-1,l$. These polynomials can be computed via the recursion

$$(l-m_l)P_l^{m_l}(x) = x(2l-1)P_{l-1}^{m_l}(x) - (l+m_l-1)P_{l-2}^{m_l}(x), \tag{II.14}$$

with starting point

$$P_{m_l}^{m_l}(x) = (-1)^{m_l}\frac{(2m_l)!}{2^{m_l}m_l!}(1-x^2)^{m_l/2}, \tag{II.15}$$

in combination with $P_l^{m_l}(x) = 0$ for $l < m_l$. Negative values of m_l are treated with the relation

$$P_l^{-m_l} = (-1)^{m_l} \frac{(l-m_l)!}{(l+m_l)!} P_l^{m_l} \,. \tag{II.16}$$

In this definition of the associated Legendre polynomial the so-called *Condon-Shortley phase* is included. This follows the definition in Ref. [6].[1] Finally, $\boldsymbol{\chi}_{s,m_s}$ in Eq. (II.12) are two-spinors defined as

$$\boldsymbol{\chi}_{\frac{1}{2},\frac{1}{2}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \ \boldsymbol{\chi}_{\frac{1}{2},-\frac{1}{2}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{II.17}$$

Explicitly (for the case of interest $s = 1/2$) the coupled basis states $\boldsymbol{\Phi}_{l,s,j,m_j}$ are:

$$\boldsymbol{\Phi}_{l,s,j,m_j} = \langle Y_{l,m_j-s} \otimes \boldsymbol{\chi}_{s,s}, \boldsymbol{\Phi}_{l,s,j,m_j} \rangle Y_{l,m_j-s} \otimes \boldsymbol{\chi}_{s,s} + \langle Y_{l,m_j+s} \otimes \boldsymbol{\chi}_{s,-s}, \boldsymbol{\Phi}_{l,s,j,m_j} \rangle Y_{l,m_j+s} \otimes \boldsymbol{\chi}_{s,-s} \,, \tag{II.18}$$

where the scalar products are (possibly vanishing) Clebsch-Gordan coefficients. These spinor transform according to irreducible representations of SU(2). This is seen in the following way: consider the unitary and faithful representation $\rho(\boldsymbol{\alpha}) = \exp(\mathrm{i}\boldsymbol{\alpha} \cdot \boldsymbol{J})$, where $\boldsymbol{\alpha} \in \mathbb{R}^3$ is a vector of length $\|\boldsymbol{\alpha}\| \in [0, 2\pi)$. Its action on the spinors is

$$\rho(\boldsymbol{\alpha})\Phi_{l,s,j,m_j}(\vartheta,\varphi) = \sum_{q=1}^{d_j} D^{(\Gamma_j)}(\boldsymbol{\alpha})_{m_j,q} \Phi_{l,s,j,q}(\vartheta,\varphi) \,. \tag{II.19}$$

In this expression, d_j is the dimension of the irreducible representation Γ_j of SU(2) and $D^{(\Gamma_j)}$ is a unitary matrix representation of the irreducible representation Γ_j.

To rewrite the Hamiltonian the crucial step is to use the fact that it has SU(2) symmetry, i.e., $[\rho(\boldsymbol{\alpha}), H] = 0$ for all $\boldsymbol{\alpha}$.[2] Due to this symmetry the Hamiltonian is automatically block diagonal with block dimensions given by integer multiplicities μ_j of the dimensions $d_j = 2j+1$ of the irreducible representations Γ_j labelled by $j = 0, 1/2, 1, 3/2, 2, \ldots$. The multiplicities μ_j are $\mu_j = 2$ for $j \neq 0$ and $\mu_0 = 0$. This is because for each $l > 0$ there are the two possible orbital angular momentum quantum numbers $l = j - 1/2$ and $l = j + 1/2$ and for $s = 1/2$ a total angular momentum of $j = 0$ is not possible. This entails that the Hamiltonian in the basis of the spinors $\boldsymbol{\Phi}_{l,s,j,m_j}$ can be decomposed into the following direct sum:

$$\bigoplus_{l=0}^{\infty} \bigoplus_{j \in \{|l-s|, l+s\}} \bigoplus_{m_j=-j}^{j} H_{l,s,j,m_j} \,, \quad H_{l,s,j,m_j} = \langle \boldsymbol{\Phi}_{l,s,j,m_j}, H \boldsymbol{\Phi}_{l,s,j,m_j} \rangle \,. \tag{II.20}$$

Note that the off-diagonal matrix elements within the blocks associated with each irreducible representation Γ_j vanish, i.e., $\langle \boldsymbol{\Phi}_{l,s,j,m_j'}, H \boldsymbol{\Phi}_{l,s,j,m_j} \rangle = 0$ for $m_j' \neq m_j$. This is a result of the $\mathrm{U}(1) \subset \mathrm{SU}(2)$ symmetry of the full Hamiltonian in Eq. (II.1), which is associated to the generator L_z. This

1 This definition entails that the factor $(-1)^{m_l}$ does not explicitly appear in the definition of the spherical harmonics. This remark is placed here to avoid confusion of whether the spherical harmonics are with or without the Condon-Shortley phase. In the definition of Y_{l,m_l} used in this work, the Condon-Shortley phase included.

2 The symmetry group is even bigger because parity is also conserved. So the full symmetry group is $\mathrm{SU}(2) \times \mathbb{Z}_2$.

symmetry further entails that the diagonal matrix elements $\langle \mathbf{\Phi}_{l,s,j,m_j}, H\mathbf{\Phi}_{l,s,j,m_j} \rangle$ do not depend on m_j. Therefore, the full Hamiltonian (II.1) in the basis of the spinors $\mathbf{\Phi}_{l,s,j}$ can be expressed as the more simple direct sum

$$\bigoplus_{l=0}^{\infty} \bigoplus_{j\in\{|l-s|,l+s\}} 1_{2j+1} \otimes H_{l,s,j} \,. \tag{II.21}$$

Here, 1_{2j+1} is the identity matrix of dimension $2j+1$ and

$$H_{l,s,j} = -\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + V_{\text{eff}}(r;\hbar^2 l[l+1], \hbar^2 s[s+1], \hbar^2 j[j+1]) \,. \tag{II.22}$$

For completeness ρ can be expressed in the basis of $\mathbf{\Phi}_{l,s,j}$ as well. It is given by the following direct sum of the irreducible representations Γ_j os SU(2):

$$\rho = \bigoplus_{j>0} 2\Gamma_j \,. \tag{II.23}$$

The block structure of the Hamiltonian reduces the eigenvalue problem defined by the time-independent Schrödinger equation

$$H\mathbf{\Psi} = E\mathbf{\Psi} \tag{II.24}$$

to the blocks $H_{l,s,j}$ on the Hilbert space associated with the radial degrees of freedom. In particular, solving for the full eigenenergies and eigenstates is reduced to solving the radial eigenvalue problems

$$H_{l,s,j}R_{n,l,s,j} = E_{n,l,s,j}R_{n,l,s,j} \,, \tag{II.25}$$

where $R_{n,l,s,j}$ is the n-th radial eigenstate of $H_{l,s,j}$. The total eigenstate $\mathbf{\Psi}$ is then given by the product of the radial eigenstates and the spinors:

$$\mathbf{\Psi}_{n,l,s,j,m_j}(r,\vartheta,\varphi) = R_{n,l,s,j}(r)\mathbf{\Phi}_{l,s,j,m_j}(\vartheta,\varphi) \,. \tag{II.26}$$

This shows that solving for the full eigenstates $\mathbf{\Psi}_{n,l,s,j,m_j}$ is equivalent to solving for the radial eigenstates $R_{n,l,s,j}$. In general, the eigenenergies $E = E_{n,l,s,j}$ and the radial eigenstates $R_{n,l,s,j}$ must be calculated numerically. This task is exactly what `AlkCalc` is designed to do.

ii. Numerical method

In this subsection the numerical method which is employed by `AlkCalc` to solve the radial eigenvalue problems in Eq. (II.26) for the eigenenergies and the radial eigenstates $R_{n,l,s,j}$ is formulated.

The first step is to express the eigenvalue problems in terms of dimensionless quantities. To this end the Hamiltonians $H_{l,s,j}$ in Eq. (II.22) are expressed in units of Hartree [6], $E_{\text{H}} = e^2/(4\pi\varepsilon_0 a_{\text{B}})$, where $a_{\text{B}} = \hbar/(m_{\text{e}}c\alpha) \approx 52.9\,\text{pm}$ is Bohr's radius. The radial eigenfunctions are parametrised as

$$R_{n,l,s,j}(r) = \frac{1}{\sqrt{a_{\text{B}}^3}}\frac{f_{n,l,s,j}(r/a_{\text{B}})}{r/a_{\text{B}}} \,, \tag{II.27}$$

where $f_{n,l,s,j}$ is dimensionless. The eigenvalue problems in Eq. (II.26) expressed in the dimensionless eigenfunctions (II.27) take the form

$$\left[-\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}t^2}+\frac{l(l+1)}{2t^2}+C\frac{V(a_\mathrm{B}t;\hbar^2 l[l+1],\hbar^2 s[s+1],\hbar^2 j[j+1])}{E_\mathrm{H}}\right]f_{n,l,s,j}(t)=\bar{E}_{n,l,s,j}f_{n,l,s,j}(t)\,, \tag{II.28}$$

where

$$C=\frac{1}{1+m_\mathrm{e}/M} \tag{II.29}$$

and the eigenenergies may be extracted from the relation

$$\frac{E_{l,s,j,n}}{E_\mathrm{H}}=\frac{\bar{E}_{n,l,s,j}}{C}\,. \tag{II.30}$$

Rewriting the normalisation condition in terms of the dimensionless eigenfunctions $f_{n,l,s,j}$ yields

$$1=\int_0^\infty \mathrm{d}r\, r^2 R^2_{n,l,s,j}(r)=\int_0^\infty (a_\mathrm{B}\,\mathrm{d}t)(a_\mathrm{B}t)^2\frac{1}{a_\mathrm{B}^3}\frac{f^2_{n,l,s,j}(t)}{t^2}=\int_0^\infty \mathrm{d}t\, f^2_{n,l,s,j}(t)\,. \tag{II.31}$$

Here it is assumed that the radial eigenfunctions are real-valued, which is justified as the Hamiltonian is not only an hermitian but also a symmetric operator. In conclusion, normalising the dimensionless eigenfunctions $f_{n,l,s,j}$ with respect to the standard one-dimensional L^2-norm automatically yields the correctly normalised radial eigenstates.

The ultimate goal is to express Eq. (II.28) as a matrix eigenvalue problem which can be directly addressed with a computer. To obtain a matrix-eigenvalue problem the first step is to choose a basis in which the dimensionless eigenfunctions can be represented. There are many different sets of basis functions to choose from. In the case of `AlkCalc` so-called *B-splines* [7] are chosen.[3] B-splines are piecewise polynomial functions with finite support. In the following it is first discussed how these B-splines, i.e., the basis functions, are constructed and subsequently the matrix eigenproblems are formulated.

The first step in constructing the B-splines is to truncate the domain of the radial eigenfunctions $f_{n,l,s,j}$ from $\mathbb{R}_{\geq 0}$ to a finite interval $I=[0,t_\mathrm{max}]\subset\mathbb{R}_{\geq 0}$, where $t_\mathrm{max}>0$ defines the maximally considered radius $r_\mathrm{max}=a_\mathrm{B}t_\mathrm{max}$. This truncation is justified whenever the maximal t_max is chosen sufficiently large such that the support of the radial eigenfunctions $f_{n,l,s,j}$ of interest lays within the interval I. Strictly speaking the support of the radial eigenstates is not finite, as they typically approach zero only asymptotically. However, in practice the support can be considered finite whenever the value of the radial eigenstates beyond a certain value of r (or equivalently t) stays below machine precision. The next step in the construction of the B-splines is to choose a family of points $(t_i)_{i=0,\ldots,N-1+2d}$ in the interval I, the so-called *knots*.[4] Here, $N\in\mathbb{N}$ is the number of

3 This method may be conceptually and programmatically somewhat more involved compared to a standard finite-difference or finite-element method, however, it is preferred here for two reasons: first, B-splines allow to work with much smaller matrix dimensions and therefore allow to efficiently solve the associated matrix-eigenvalue problems on a computer. Second, the small matrix dimensions allow to space-efficiently store the computed radial eigenstates.

4 The family $(t_i)_{i=0,\ldots,N-1+2d}$ is sometimes called the *knot vector* in the literature.

unique knots and $d \geq 0$ is the degree of the B-splines: each B-spline is a piecewise polynomial function; a degree-d B-spline is a B-spline where all these polynomials have (maximally) degree d. The knots t_i are defined as

$$t_i = \begin{cases} 0 & , \quad 0 \leq i \leq k-1 \\ t_i & , \quad k \leq i \leq k+N-3 \\ t_{\max} & , \quad k+N-2 \leq i \leq 2k+N-3 \end{cases} , \tag{II.32}$$

where the so-called $N-2$ *internal knots* $t_k < \cdots < t_{k+N-3}$ are chosen in strictly ascending order. This construction can be visualised as follows:

$$\underbrace{t_0, \ldots, t_0}_{k \text{ times}}, \underbrace{t_k, \ldots, t_{k+N-3}}_{\text{Internal knots}}, \underbrace{t_{\max}, \ldots, t_{\max}}_{k \text{ times}} . \tag{II.33}$$

There are N different knots. The first and the last one are repeated $k = d+1$ times, where k is called the *order* of the B-splines.

On the knots $(t_i)_{i=0,\ldots,N-1+2d}$ the $N_{\mathrm{bs}} = N + d - 1$ many B-splines $u_i^{(d)}$ of degree d (or order k) are defined via de Boor's recursion formula [7] ($q = 1, \ldots, d$, $i = 0, \ldots, N_{\mathrm{bs}} - 1$):

$$u_i^{(q)}(t) = \frac{t - t_i}{t_{i+q} - t_i} u_i^{(q-1)}(t) + \frac{t_{i+q+1} - t}{t_{i+q+1} - t_{i+1}} u_{i+1}^{(q-1)}(t) , \quad u_i^{(0)}(t) = \begin{cases} 1 & , \quad t_i \leq t < t_{i+1} \\ 0 & , \quad \text{else} \end{cases} . \tag{II.34}$$

The support of the resulting B-splines is restricted to the interval $[t_i, t_{i+k})$. One can show that with the chosen initial condition in Eq. (II.34) all B-splines $u_i^{(d)} : \mathbb{R} \to [0,1]$, i.e., map into the interval $[0,1]$.[5] Further one shows that

$$\sum_{i=0}^{N_{\mathrm{bs}}-1} u_i^{(d)} = 1 . \tag{II.35}$$

These properties entail that the basis of B-splines

$$\left(u_0^{(d)}, \ldots, u_{N_{\mathrm{bs}}-1}^{(d)} \right) \tag{II.36}$$

is a partition of unity. With this normalisation the B-spline basis is uniquely defined for the choice of knots $(t_i)_{i=0,\ldots,N-1+2d}$. With this the construction of the B-spline basis is constructed. The next step is to express the eigenvalue problem posed by Eq. (II.28) in this basis.

With the B-spline basis at hand, the next step is to express the eigenvalue problem posed by Eq. (II.28) in that basis. To this end one integrates over the eigenvalue problem with a test function g to reduce it to its so-called *weak form*:

$$\int_{-\infty}^{\infty} \mathrm{d}t \, g(t) \left(-\frac{1}{2} \frac{\mathrm{d}^2}{\mathrm{d}t^2} + v(t) \right) f(t) = \lambda \int_{-\infty}^{\infty} \mathrm{d}t \, g(t) f(t) . \tag{II.37}$$

5 The B-splines are defined on the full set $\mathbb{R}$ with the understanding that each B-spline is zero outside the interval I.

To simplify the notation the indices on $f_{n,l,s,j}$ were dropped and the dimensionless energy renamed to $\lambda = \bar{E}_{n,l,s,j}$. Further, the potential terms in Eq. (II.28) are collected into the function

$$v(t) = \frac{l(l+1)}{2t^2} + C\frac{V(a_\mathrm{B}t; \hbar^2 l[l+1], \hbar^2 s[s+1], \hbar^2 j[j+1])}{E_\mathrm{H}}\,. \tag{II.38}$$

To reduce the weak form of the eigenvalue problem to a matrix eigenvalue problem one expands the eigenfunctions f in the B-spline basis:

$$f \approx \sum_{i=0}^{N_\mathrm{bs}-1} f_i u_i^{(d)}\,. \tag{II.39}$$

Here, $\boldsymbol{f} = (f_1, \dots, f_{N_\mathrm{bs}-1})^\mathrm{T}$ is the vector containing the expansion coefficients f_i. It is important to note that this is an approximation, because in general one must employ an infinite basis to exactly represent the eigenfunctions f. A brief discussion regarding this approximation is reserved for later in this section. For now, the expansion of the eigenfunctions f given in Eq. (II.39) can be plugged into Eq. (II.37). With the choice $g = u_i^{(d)}$ one arrives at the expression

$$\sum_{j=0}^{N_\mathrm{bs}-1} f_j B\left(u_i^{(d)}(t), -\frac{1}{2}(u_j^{(d)})''(t) + v(t)u_j^{(d)}(t) \right) = \lambda \sum_{i=0}^{N_\mathrm{bs}-1} f_j B(u_i^{(d)}(t), u_j^{(d)}(t))\,. \tag{II.40}$$

Here, for convenience, the bilinear form

$$B(\alpha, \beta) = \int_I \mathrm{d}t\, \alpha(t)\beta(t) \tag{II.41}$$

was introduced for any two sufficiently well-behaved functions $\alpha, \beta : I \to \mathbb{R}$.

Next, the homogeneous Dirichlet boundary conditions $f(0) = f(t_\mathrm{max}) = 0$ are enforced on f in Eq. (II.39). To this end one notices that for the choice of knots employed in this work the de Boor algorithm [7] is defined such that $u_i^{(d)}(0) = \delta_{i,0}$ and $u_i^{(d)}(0) = \delta_{i,N_\mathrm{bs}-1}$. This entails that $f(0) = f_0$ and $f(t_\mathrm{max}) = f_{N_\mathrm{bs}-1}$. As a result, the boundary conditions are uniquely enforced by $f_0 = f_{N_\mathrm{bs}-1} = 0$. Employing this result the sums in Eq. (II.40) is be reduced by the boundary terms:

$$\sum_{j=1}^{N_\mathrm{bs}-2} f_j B\left(u_i^{(d)}(t), -\frac{1}{2}(u_j^{(d)})''(t) + v(t)u_j^{(d)}(t) \right) = \lambda \sum_{i=1}^{N_\mathrm{bs}-2} f_j B(u_i^{(d)}(t), u_j^{(d)}(t))\,. \tag{II.42}$$

To finally bring the eigenvalue problem posed by Eq. (II.28) into matrix form, the following matrices shall be defined: first, the so-called *stiffness* matrix K, which is a $(N_\mathrm{bs}-2) \times (N_\mathrm{bs}-2)$-dimensional matrix with components ($i, j = 1, \dots, N_\mathrm{bs}-2$) defined as

$$K_{i,j} = -\frac{1}{2}B((u_i^{(d)})'', (u_j^{(d)})) = \frac{1}{2}B((u_i^{(d)})', (u_j^{(d)})')\,. \tag{II.43}$$

The second equal sign is obtained by partial integration with the fact that for all indices i, j the B-splines obey $u_{i(j)}^{(d)}(0) = u_{i(j)}^{(d)}(t_\mathrm{max}) = 0$. The next matrix is the so-called *mass* matrix M.[6] This

6 The terminology 'stiffness' and 'mass' matrix is common terminology in literature on the finite-element method.

$(N_{\mathrm{bs}}-2)\times(N_{\mathrm{bs}}-2)$-dimensional matrix accounts for the fact that the B-splines are not orthogonal with respect to the bilinear form B in Eq. (II.41). Its components ($i, j = 1, \ldots, N_{\mathrm{bs}} - 2$) are defined as

$$M_{i,j} = B(u_i^{(d)}, u_j^{(d)}) \,. \tag{II.44}$$

The last matrix W represents the potential. The components of this $(N_{\mathrm{bs}}-2)\times(N_{\mathrm{bs}}-2)$-dimensional matrix ($i, j = 1, \ldots, N_{\mathrm{bs}} - 2$) are defined as

$$W_{i,j} = B(u_i^{(d)}, v u_j^{(d)}) \,. \tag{II.45}$$

Importantly, all three matrices K, M, and W are real, symmetric, and banded. The banded structure is a consequence of the finite support of the B-splines. Combined, these properties allow to employ efficient specialised numerical methods to solve for the coefficient vectors $\boldsymbol{f}$ and the eigenvalues λ.

Expressed in the three matrices K, M, and W the eigenvalue problem posed by Eq. (II.42) takes the form

$$(K + W)\bar{\boldsymbol{f}} = \lambda M \bar{\boldsymbol{f}} \,, \tag{II.46}$$

where $\bar{\boldsymbol{f}} = (f_1, \ldots, f_{N_{\mathrm{bs}}-2})^{\mathrm{T}}$ is the $(N_{\mathrm{bs}} - 2)$-dimensional coefficient vector which defines the full coefficient vector $\boldsymbol{f} = (0, \bar{\boldsymbol{f}}^{\mathrm{T}}, 0)^{\mathrm{T}}$ including the coefficients $f_1 = f_{N_{\mathrm{bs}}-1} = 0$. Therefore, a solution to the (generalised) matrix eigenvalue problem problem posed by Eq. (II.46) is equivalent to obtaining the full coefficient vector $\boldsymbol{f}$ and thereby an approximate solution to the eigenfunctions of the exact eigenvalue problem posed by Eq. (II.28). Note that the conversion formula in Eq.+(II.27) finally relates this result directly to the radial eigenstates $R_{n,l,s,j}$. The normalisation of the eigenfunctions f is trivially achieved by normalising the coefficient vector with respect to the scalar product B represented by the positive-definite matrix M, i.e., in order to normalise f one must normalise $\bar{\boldsymbol{f}}$ such that

$$1 = \bar{\boldsymbol{f}}^{\mathrm{T}} M \bar{\boldsymbol{f}} \,. \tag{II.47}$$

This can be seen directly form the normalisation condition in Eq. (II.31) combined with the following calculation

$$1 = \int_{-\infty}^{\infty} \mathrm{d}t\, f^2(t) \approx \sum_{i=1}^{N_{\mathrm{bs}}-2} \sum_{j=1}^{N_{\mathrm{bs}}-2} f_i f_j B(u_i^{(d)}, u_j^{(d)}) = \sum_{i=1}^{N_{\mathrm{bs}}-2} \sum_{j=1}^{N_{\mathrm{bs}}-2} f_i M_{i,j} f_j = \bar{\boldsymbol{f}}^{\mathrm{T}} M \bar{\boldsymbol{f}} \,. \tag{II.48}$$

Finally, to solve the generalised eigenvalue problem posed by Eq. (II.46) `AlkCalc` employs the routine ??? from the software collection LAPACK [**LAPACK**].

At this point there are two more open questions to answer. **(i)** The (generalised) matrix eigenvalue problem posed by Eq. (II.46) is an approximation to the exact eigenvalue problem posed by Eq. (II.28). One might ask if the numerically computed eigenvalues and eigenvectors indeed approximate the exact result. For instance, on glimpse one might think that it is possible to find eigenvalues which are not physical but solely numerical artefacts of the approximate expansion of f in a finite set of basis states. **(ii)** The second question is how the components of the matrices K, M, and W can be calculated numerically in an efficient manner, as these are given by integrals that, generally, need to be solved numerically.

(i) Ritz-variational methods for argumentation

(ii) Gaussian quadratures

iii. Radial matrix elements

To compute radial matrix elements one calculates, for $p \in \mathbb{R}$ such that the integral exists,

$$\begin{aligned}
\frac{r_p^{n,l,s,j;n',l',s',j'}}{a_\mathrm{B}^p} &\equiv \frac{1}{a_\mathrm{B}^p}\langle R_{n,l,s,j}, \mathrm{id}^p R_{n',l',s',j'}\rangle = \int_0^\infty \mathrm{d}r\, r^2 R_{n,l,s,j}(r)\frac{r^p}{a_\mathrm{B}^p}R_{n',l',s',j'}(r) \\
&= \int_0^\infty (\mathrm{d}t\, a_\mathrm{B})(a_\mathrm{B}t)^2 f_{n,l,s,j}(t)\frac{(a_\mathrm{B}t)^p}{a_\mathrm{B}^p}\frac{1}{a_\mathrm{B}^3}\frac{f_{n',l',s',j'}(t)}{t^2} \\
&\approx \int_I (\mathrm{d}t\, a_\mathrm{B})(a_\mathrm{B}t)^2 f_{n,l,s,j}(t)\frac{(a_\mathrm{B}t)^p}{a_\mathrm{B}^p}\frac{1}{a_\mathrm{B}^3}\frac{f_{n',l',s',j'}(t)}{t^2} = B(f_{n,l,s,j}, \mathrm{id}^p f_{n',l',s',j'})\,.
\end{aligned} \tag{II.49}$$

Here, 'id' denotes the identity map. The remaining scalar product can be approximated with solutions of the matrix eigenproblem in Eq. (??), i.e., $f_{n,l,s,j}$ and $f_{n',l',s',j'}$ are represented by the vectors $\boldsymbol{f}_{n,l,s,j}$ and $\boldsymbol{f}_{n',l',s',j'}$, respectively [see Eq. (??)]. Within this approximations

$$B(f_{n,l,s,j}, \mathrm{id}^p f_{n',l',s',j'}) = \boldsymbol{f}_{n,l,s,j}^\mathrm{T} R_p \boldsymbol{f}_{n',l',s',j'}\,, \tag{II.50}$$

where the components of the real symmetric matrix R_p are defined as

$$(R_p)_{k,k'} = B(\phi_k, \mathrm{id}^p\phi_{k'}) = \begin{cases} B(\phi_k, \mathrm{id}^p\phi_{k+1}), & k' = k+1 \text{ or } k = k'+1 \ (0 < k, k' < N-2) \\ B(\phi_k, \mathrm{id}^p\phi_k), & k' = k \ (0 < k, k' \leq N-2) \\ 0, & \text{else} \end{cases}\,. \tag{II.51}$$

To numerically compute the remaining integrals we use a two-point Gaußian quadrature. This method approximates the integral as [8]:

$$\int_a^b \mathrm{d}t\, \alpha(t) \approx \frac{b-a}{2}\left[\alpha\left(\frac{a+b}{2} - \frac{b-a}{2\sqrt{3}}\right) + \alpha\left(\frac{a+b}{2} + \frac{b-a}{2\sqrt{3}}\right)\right], \tag{II.52}$$

for any sufficiently smooth function α and $a < b$. Note that this result is exact if α is a polynomial of degree three, which is the case for radial dipole matrix elements. This method is straightforward to implement in a computer program, such that the problem of calculating matrix elements is effectively reduced to computing matrix products. However, note that for too large p there can be numerical overflow or underflow when computing t^p. But, for 64-bit floating point arithmetic, $p = \pm 10$ is easily possible (if the integral exists), such that, typically, all practical cases are included. Explicitly, we could verify that for Hydrogen the range $p = -2, \ldots, 10$ yields reasonably good results. Finally, we remark that one can improve on the precision by employing higher-order Gaußian quadrature algorithms as described in [8].

iv. Oscillator strengths

The so-called *oscillator strength* of a transition from an initial state $\boldsymbol{\Psi}_\mathrm{i} \equiv \boldsymbol{\Psi}_{n',l',s,j',m_j'}$ to a final state $\boldsymbol{\Psi}_\mathrm{f} \equiv \boldsymbol{\Psi}_{n,l,s,j,m_j}$ is [6]

$$f_{\mathrm{i}\to\mathrm{f}} = \sum_{m_j'=-j'}^{j'} \frac{2}{3}\bar{E}_{\mathrm{f,i}}\left|\langle\boldsymbol{\Psi}_\mathrm{f}, \boldsymbol{r}/a_\mathrm{B}\boldsymbol{\Psi}_\mathrm{i}\rangle\right|^2, \tag{II.53}$$

with the energy difference $E_\mathrm{H}\bar{E}_{\mathrm{f,i}} = E_\mathrm{f} - E_\mathrm{i}$. Note that the sum runs over all final m_j' because one is interested in transitions between the fine-structure states without selecting Zeeman sublevels. The oscillator strength is a measure for how strongly the induced Hertzian (electric) dipole oscillates: the radiation power is large for spatially large dipoles and fast oscillation frequencies, captured by the matrix element and energy difference, respectively. In principle, all the machinery to numerically compute this quantity is already there, however, one can reduce the expression to a much simpler form by using angular momentum algebra, in particular, relations from [9]. To simplify the expression, first note (Condon-Shortley phase convention)

$$r_x = \sqrt{\frac{2\pi}{3}} r(Y_{1,-1} - Y_{1,1})\,,\ r_y = \mathrm{i}\sqrt{\frac{2\pi}{3}} r(Y_{1,-1} + Y_{1,1})\,,\ r_z = \sqrt{\frac{4\pi}{3}} r Y_{1,0}\,. \tag{II.54}$$

With this, the oscillator strength becomes

$$f_{\mathrm{i}\to\mathrm{f}} = \frac{4\pi}{3}\frac{2}{3}\bar{E}_{\mathrm{f,i}}\langle rR_\mathrm{f}, r^2/a_\mathrm{B} rR_\mathrm{i}\rangle^2 \sum_{m_j'=-j'}^{j'} \sum_{q=-1}^{1} |\langle \Phi_\mathrm{f}, Y_{1,q}\Phi_\mathrm{i}\rangle|^2 \tag{II.55}$$

Employing relations from [9, p. 236, 244, 260, 445, 496] one can show that the remaining matrix element is

$$\langle \Phi_\mathrm{f}, Y_{1,q}\Phi_\mathrm{i}\rangle = (-1)^{j'+j+1+l'+s-m_j'} \sqrt{\frac{3}{4\pi}} \begin{pmatrix} j' & 1 & j \\ -m_j' & q & m_j \end{pmatrix} \sqrt{(2j+1)(2j'+1)(2l+1)} \begin{Bmatrix} l & s & j \\ j' & 1 & l' \end{Bmatrix} C^{l,0,1,0}_{l',0}\,, \tag{II.56}$$

expressed in terms of a Wigner $3jm$-symbol, a Wigner $6j$-symbol, and a Clebsch-Gordan coefficient. With symmetry relations and a sum rule from [9, p. 236, 259] one can compute the sum over q and m_j' in the expression for the oscillator strength, which yields

$$\sum_{m_j'} f_{\mathrm{i}\to\mathrm{f}} = \frac{2}{3}\bar{E}_{\mathrm{f,i}} |\langle rR_{n',l',s,j'}, r^2/a_\mathrm{B} R_{n,l,s,j}\rangle|^2 a_l\,,\ a_l = l_\mathrm{max}(2j'+1)\begin{Bmatrix} l & s & j \\ j' & 1 & l' \end{Bmatrix}^2\,, \tag{II.57}$$

where $l_\mathrm{max} = \max(l, l')$. There are only six transitions for which a_l is non-zero. Using relations from [9, p. 298, 300] one arrives at the results shown in Tab. 1. With this it is now straight forward to compute the oscillator strengths numerically with a computer program. As a final remark, note that the oscillator strength does not depend on m_j any more, i.e., after summing over final states, the oscillators for different m_j all have the same strength.

v. State lifetimes

The lifetime τ of a fine-structure state is given by the formula

$$\tau = \Gamma^{-1}\,, \tag{II.58}$$

where Γ is the total rate at which the state is depopulated. At sufficiently low temperatures this rate is given by the sum over the so-called *Einstein-A* coefficients $A_{\mathrm{i}\to\mathrm{f}}$ [6, 10]. These coefficients are the rates at which the state of interest, here referred to as the 'initial state', radiatively decays

(l,j)	$\mapsto$	(l',j')	a_l
$(l,l+s)$	$\mapsto$	$(l+1,l+s)$	$\frac{1}{(2l+3)(2l+1)}$
$(l,l-s)$	$\mapsto$	$(l+1,l+s)$	$\frac{l+1}{2l+1}$
$(l,l+s)$	$\mapsto$	$(l+1,l+3s)$	$\frac{l+2}{2l+3}$
$(l,l-s)$	$\mapsto$	$(l-1,l-s)$	$\frac{1}{(2l+1)(2l-1)}$
$(l,l+s)$	$\mapsto$	$(l-1,l-s)$	$\frac{l}{2l+1}$
$(l,l-s)$	$\mapsto$	$(l-1,l-3s)$	$\frac{l-1}{2l-1}$

Table 1: Angular factors. Table containing the angular factors a_l [see Eq. (II.57)] for computing oscillator strengths. Throughout this table, $s = 1/2$.

to some final state. In the dipole approximation the sum over Einstein-A coefficients can be expressed as [6]:

$$A_{\mathrm{i}} = \sum_{\mathrm{f}} A_{\mathrm{i}\to\mathrm{f}}\,,\; A_{\mathrm{i}\to\mathrm{f}} = \frac{E_{\mathrm{H}}}{\hbar}\frac{4}{3}\alpha^2 \bar{E}_{\mathrm{i,f}}^3 \sum_{u=x,y,z} |\langle \boldsymbol{\Psi}_{\mathrm{f}}, r_u/a_{\mathrm{B}} \boldsymbol{\Psi}_{\mathrm{i}}\rangle|^2\,. \tag{II.59}$$

In terms of oscillator strengths one obtains

$$A_{\mathrm{i}} = \frac{2\alpha^3 E_{\mathrm{H}}}{\hbar} \sum_{n',l',j'} \bar{E}_{\mathrm{i,f}}^2 (-f_{\mathrm{i}\to\mathrm{f}})\,. \tag{II.60}$$

The rate A_{i} in Eq. (II.60) is the spontaneous emission rate, which is only non-zero for transitions where a photon is emitted and the final state's energy is lower than the energy of the initial state, i.e., $E_{\mathrm{i}} > E_{\mathrm{f}}$. Note that this restricts the quantum numbers in the sum.

At sufficiently high temperatures the spectral density of the environment cannot be neglected. To approximate the spectral density of the environment one can employ Plank's famous formula [11] for the spectral density of a black-body. This yields an average number of photons at frequency ω and temperature T:

$$n(\omega, T) = \frac{1}{\exp(\hbar\omega/(k_{\mathrm{B}}T)) - 1}\,. \tag{II.61}$$

Here, k_{B} is Boltzmann's constant. Compared to the low temperature regime, the additional photons at temperature T stimulate the emission of photons and thereby the decay of the initial state to lower-energy final states. Even depopulation of the initial state is possible by the absorption of photons from the environment. This process drives transitions to final states which are higher in energy than the initial state. According to Einstein's rate equations [10] the total rate for depopulation of the initial state is

$$\frac{\hbar}{E_{\mathrm{H}}}\Gamma = 2\alpha^3 \sum_{n',l',j'} \bar{E}_{\mathrm{i,f}}^2(-f_{\mathrm{i}\to\mathrm{f}})(1 + n(E_{\mathrm{i,f}}/\hbar, T)) + 2\alpha^3 \sum_{n',l',j'} \bar{E}_{\mathrm{f,i}}^2 f_{\mathrm{i}\to\mathrm{f}}\, n(E_{\mathrm{f,i}}/\hbar, T)\,, \tag{II.62}$$

where each sum is restricted to quantum numbers for which each summand is positive. This entails that the first sum runs only over final states with lower energy compared to the initial state; this describes the emission of photons. The second sum runs solely over final states with larger energy than the initial state; this corresponds to the case of photon absorption.

III. Manual

Reference manual for the library functions of `AlkCalc`. Prototypes for the following functions are defined in the header 'interface/alkcalc.h'.

There is always a decision to be made when dealing with half-integer quantum numbers numerically. Here the following agreement is made: say q is a quantum number that may take half-integer values, e.g., j, m_j. To 'clean up' the numerical value it is understood that $2q$ is equivalent to $[2.0 \times q + 0.5]$. The 'floor' function is defined as $[x] = k \in \mathbb{Z}$ ($x \in \mathbb{R}$), where k is the unique integer such that $x \in [k, k+1)$. In practice, this means that for any $q \in [-0.75, -0.25)$, $q = -1/2$ is used, for any $q \in [-0.25, 0.25)$, $q = 0$, for any $q \in [0.25, 0.75)$, $q = 1/2$, and so on, in both the negative and positive directions. In the following manual, this convention is applied to all quantum numbers that can take half-integer values, unless stated otherwise.

FUNCTION

`alkcalc_Enlsj`

DESCRIPTION

Eigenenergies in units of Hartree. The requested eigenenergy is read from the corresponding data file in the directory 'data' and returned.

ARGUMENTS

`char *species`

String to specify atom/ion species, e.g., 1H for Hydrogen, or `88SR+` for the $^{88}Sr^{+}$ ion.

`int32_t n`

Principal quantum number.

`int32_t l`

Orbital angular momentum quantum number $l = 0, 1, \dots, n-1$.

`double s`

Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double j`

Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$.

RETURN

`double E`

E is the eigenenergy specified by the quantum numbers n, l, $s = 1/2$, and j.

FUNCTION

`alkcalc_fnlsj`

DESCRIPTION

Numerical equivalent to $f_{n,l,s,j}$ in Eq. (??). The data is read from the file containing the requested state. This data is stored at a user specified location. The location where `AlkCalc` will search for the data can be set in the file 'interface/alkcalc.h'. The result is owned by the caller and should be freed with `alkcalc_state_free`.

ARGUMENTS

`char result`

Character for deciding if discretisation data is returned; see RETURN down below. If `result` is `f` (full), the full result is returned, and if `result` is `p` (partial), the discretisation data is not returned.

`char *species`

String to specify atom/ion species, e.g., `1H` for Hydrogen, or `88SR+` for the $^{88}Sr^{+}$ ion.

`int32_t n`

Principal quantum number.

`int32_t l`

Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$.

`double s`

Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double j`

Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$.

RETURN

`alkcalc_state *f`

`f` contains eight members: `N`, `dim`, `t`, `h`, and `fnlsj`.

- `int32_t N`: Number of points, N, to discretise the interval $[0, t_{\max}]$ (see Sec. II).
- `int32_t n`: Principal quantum number.
- `int32_t l`: Orbital angular momentum quantum number.
- `double j`: Total angular momentum quantum number.
- `int32_t dim`: Dimension of eigenproblem, i.e., `dim` $= N - 2$ [see Eq. (??)].
- `double *t`: Array of length N, containing the mesh $(t_k)_{k=0,\ldots,N-1}$ (see Sec. II).
- `double *h`: Array of length $N-1$, containing the steps $(h_k)_{k=1,\ldots,N-1}$ (see Sec. II).
- `double *fnlsj`: `dim`-dimensional eigenvector f [see Eq. (??)].

FUNCTION

`alkcalc_rp`

DESCRIPTION

Radial matrix element of r^p, $p \in \mathbb{R}$ [see Eq. (II.49)]. For $|p| \gtrsim 10$ there is the risk of numerical over or underflow of the 64-bit floating-point arithmetic. It is the user's responsibility to ensure that the integral associated to the desired matrix element exists. In case it does not, the returned value is wrong and no error is thrown. ^{1}H (Hydrogen atom) and ^{2}He$^+$ (Helium ion) can be used to benchmark atoms and ions, respectively.

ARGUMENTS

`char *species`
String to specify atom/ion species, e.g., 1H for Hydrogen, or 88SR+ for the ^{88}Sr$^+$ ion.

`int32_t nb`
Principal quantum number of bra.

`int32_t lb`
Orbital angular momentum quantum number $l = 0, 1, \dots, n-1$ of bra.

`double sb`
Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double jb`
Total angular momentum quantum number of bra, j.

`double p`
Power of radius operator; cf. Eq. (II.49). For instance, $p = 0$ yields the scalar product of the states. For $p = 1$ and $p = 2$, respectively, dipole and quadrupole matrix elements are computed. Generally, $p \in \mathbb{R}$.

`int32_t nk`
Principal quantum number of ket.

`int32_t lk`
Orbital angular momentum quantum number $l = 0, 1, \dots, n-1$ of ket.

`double sk`
Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double jk`
Total angular momentum quantum number of ket, j.

RETURN

`double rp`
$r_p^{n,l,s,j;n',l',s',j'}$ in Eq. (II.49) in units of a_B^p, where a_B is Bohr's radius. Quantum numbers with a prime correspond to the bra, and the ones without one, to the ket.

FUNCTION

`alkcalc_cj1m1j2m2jmj`

DESCRIPTION

Clebsch-Gordan coefficients, $C^{j_1,m_1,j_2,m_2}_{j,m_j} = \langle j_1, m_1; j_2, m_2 | j_1, j_2, j, m_j \rangle$, where j is the total angular momentum composed of the two individual ones j_1 and j_2. To fix the phase uniquely, the Condon-Shortley sign convention is adopted, that is, the Clebsch-Gordan coefficients with $m_j = j_1 + j_2$ are positive. In particular, for the case $j_1 = l$ and $j_2 = s$,

$$C^{l,m_l,s,m_s}_{j,m_j} = \left\langle Y_{l,m_l} \otimes \boldsymbol{\chi}_{s,m_s}, \Phi_{l,s,j,m_j} \right\rangle , \tag{III.1}$$

with the states $Y_{l,m_l} \otimes \boldsymbol{\chi}_{s,m_s}$ of the uncoupled basis and the coupled basis states $\Phi_{l,s,j,m}$; see Eqs. (II.13, II.17), and Eq. (II.18), respectively. The return type (see RETURN down below) allows to extract the requested coefficient symbolically as well as numerically. The function returns zero in case the arguments do not make sense, e.g., if $j_1 < m_1$; or if j_1 is half-integer and m_1 is integer; or if j does not lay between (including bounds) $|j_1 - j_2|$ and $|j_1 + j_2|$.

ARGUMENTS

`double j1`
Angular momentum quantum number of first factor.

`double m1`
Magnetic quantum number of first factor.

`double j2`
Angular momentum quantum number of second factor.

`double m2`
Magnetic quantum number of second factor.

`double j`
Total angular momentum quantum number.

`double mj`
Magnetic quantum number of total angular momentum.

RETURN

`alkcalc_cg c`
c contains three members: `sign`, `numerator`, and `denominator`.
`int8_t sign`: Sign, σ, of the Clebsch-Gordan coefficient.
`int64_t numerator`: Numerator, ν, of the Clebsch-Gordan coefficient.
`int64_t denominator`: Denominator, μ, of the Clebsch-Gordan coefficient.
The requested Clebsch-Gordan coefficient is given by $\sigma\sqrt{\nu/\mu}$. Euklid's algrithm is employed to reduce ν and μ. This representation allows to obtain Clebsch-Gordan coefficients symbolically as well as numerically.

FUNCTION

`alkcalc_YlmlXsms`

DESCRIPTION

Angular factor of the full state in the uncoupled basis, i.e., the two-component spinor $Y_{l,m_l}(\vartheta,\varphi)\chi_{s,ms}$. The input quantum numbers are checked for inconsistencies and an error is thrown in case one is detected. The angles ϑ and φ have the ranges $[0,\pi]$ and $[0,2\pi]$, respectively. This condition is checked up to floating point machine precision (type `double`) and an error will be thrown in case either condition is not obeyed.

ARGUMENTS

`int32_t l`
: Orbital angular momentum quantum number $l = 0, 1, \dots, n-1$.

`int32_t ml`
: Magnetic quantum number of orbital angular momentum.

`double s`
: Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double ms`
: Magnetic quantum number of electron spin, m_s. For any $m_s < 0$ the value $m_s = -1/2$ is used internally, and for $m_s \geq 0$ the value $m_s = 1/2$.

`double theta`
: Polar angle (Zenitwinkel), ϑ, in range $[0,\pi]$.

`double phi`
: Azimuthal angle (Azimut), φ, in range $[0,2\pi]$.

RETURN

`alkcalc_spinor s`
: `s` has two components: `u` and `v`.
  `u` is the spin-up ($m_s = 1/2$) component.
  `d` is the spin-down ($m_s = -1/2$) component.

FUNCTION

`alkcalc_Philsjmj`

DESCRIPTION

Angular factor of the full eigenstate in the coupled (fine-structure) basis, i.e., the two-component spinor $\Phi_{l,s,j,mj}(\vartheta, \varphi)$ [see Eq. (II.18)]. The input quantum numbers are checked for inconsistencies and an error is thrown in case one is detected. The angles ϑ and φ have the ranges $[0, \pi]$ and $[0, 2\pi]$, respectively. This condition is checked up to floating point machine precision (type `double`) and an error will be thrown in case either condition is not obeyed. To compute the resulting two-component spinor, Clebsch-Gordon coefficients are needed, for which the sign convention detailed in Eq. (III.1) is employed.

ARGUMENTS

`int32_t l`
: Orbital angular momentum quantum number $l = 0, 1, \dots, n-1$.

`double s`
: Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double j`
: Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$.

`double mj`
: Magnetic quantum number of total angular momentum, m_j.

`double theta`
: Polar angle (Zenitwinkel), ϑ, in range $[0, \pi]$.

`double phi`
: Azimuthal angle (Azimut), φ, in range $[0, 2\pi]$.

RETURN

`alkcalc_spinor s`
: `s` has two components: `u` and `v`.
  `u` is the spin-up ($m_s = 1/2$) component.
  `d` is the spin-down ($m_s = -1/2$) component.

FUNCTION

`alkcalc_fitof`

DESCRIPTION

Oscillator strength for the transition between an initial state Ψ_i and a final state Ψ_f, as defined in Subsec. iv, that is, it was already summed over the magnetic quantum number of the final state.

ARGUMENTS

`char *species`
String to specify atom/ion species, e.g., 1H for Hydrogen, or 88SR+ for the $^{88}Sr^{+}$ ion.

`int32_t ni`
Principal quantum number of the initial state Ψ_i.

`int32_t li`
Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$ of the initial state Ψ_i.

`double si`
Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double ji`
Total angular momentum quantum number of the initial state Ψ_i, j.

`int32_t nf`
Principal quantum number of the final state Ψ_f.

`int32_t lf`
Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$ of the final state Ψ_f.

`double sf`
Electron spin quantum number of the final state Ψ_f. Not an argument here, since $s = 1/2$ is always assumed.

`double jf`
Total angular momentum quantum number of the final state Ψ_f, j.

RETURN

`double fitof`
$f_{\mathrm{i}\to\mathrm{f}}$ is the oscillator strength of the selected electronic transition, summed over m_j' of the final state. Note that because of this sum, it is the same for all m_j of the initial state.

FUNCTION

`alkcalc_tau`

DESCRIPTION

Lifetime of the fine-structure state Ψ_{n,l,s,j,m_j} in nanoseconds. Note that, since the oscillator-strengths [see Eq. (II.57)] do not depend on m_j, also the lifetimes do not depend on it [see Eq. (II.62)].

ARGUMENTS

`double T`

Temperature of black-body excitation spectrum in Kelvin.

`char *species`

String to specify atom/ion species, e.g., 1H for Hydrogen, or 88SR+ for the $^{88}Sr^{+}$ ion.

`int32_t n`

Principal quantum number.

`int32_t dn`

Range for principal quantum number. At zero temperature, this argument is not used. For finite temperature, the absorption processes in Eq. (II.62) allow 'decay' to states with higher energy. In principle, one needs to take into account all transitions to these states, of which there are infinitely many. Here, solely dn levels above Ψ_{n,l,s,j,m_j} are taken into account, in order to truncate the series, for both $l' = l \pm 1$. In other words, the sum in Eq. (II.62), taking into account absorption, contains $2 \times$ dn terms for $l > 0$ and dn for $l = 0$.

`int32_t l`

Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$.

`double s`

Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double j`

Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$.

RETURN

`double tau`

τ is the lifetime of the fine-structure state Ψ_{n,l,s,j,m_j} (independent of m_j) in nanoseconds.

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